

# GATI

ISO  
20121:2012



SREENIDHI'S ANNUAL TECHNICAL FEST

★ ISO 20121:2012 CERTIFIED

# ROBOVEDA'24

EXPLORE THE UNWRITTEN LORE...

# R U L E B O O K

## GATI

### 1. GAME AND RULES:

#### 1.1. PROBLEM STATEMENT

1.1.1. Design a wired or wireless robot with the specified dimensions that can travel through all turns of the track. The robot that cracks the arena in the least amount of time is the winner.

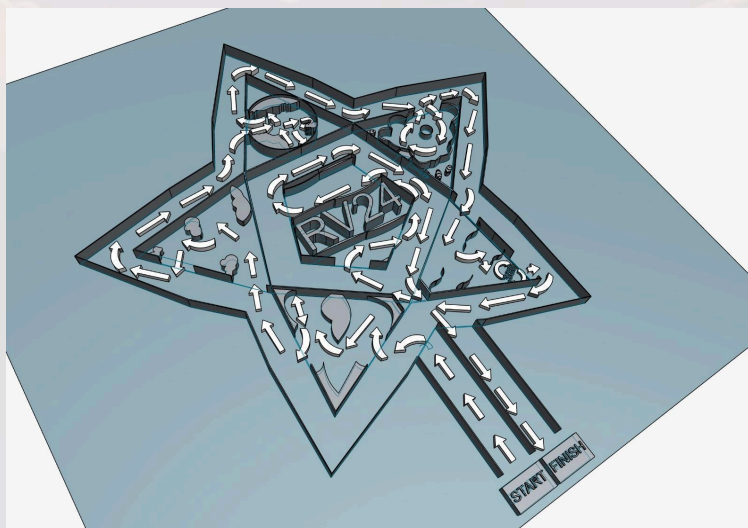
### 2. GAME FIELD AND OBJECTS :

#### 2.1. ARENA

2.1.1. The arena is placed on the ground, there may be obstacles at certain points in the racecourse.

2.1.2. Minimum track width of the arena is 30cm.

Note: The design is not drawn to scale. The orthographic view of the arena is shown below.



### 3. TEAMS:

3.1. There is no team limit.

3.2. Only 2 members of a team are allowed to stay around the arena (one for controlling and one for assisting) and a team is allowed to play with one robot only.

3.3. Members of different institutions can form a team.

3.4. Students who are pursuing any course upto post graduation can participate.



## 4. ROBOTS:

### 4.1. ROBOT SPECIFICATIONS

4.1.1. Robots can be wired or wireless.

4.1.2. Robots should not damage any part of the arena. If the robot damages any part of the arena, it will be disqualified.

4.1.3. The robot should not expand or compress during the run. A technical timeout can be taken in case of any part detachment of the robot.

4.1.4. Participants are allowed to participate only once in that particular event. A robot is allowed to participate only once in the event. The robot can participate once again with the same team if it satisfies the following conditions:

- The team has to register once again with a modified old robot/new robot before registrations close and a maximum of 3 registrations will be accepted.
- Wheels or motors or chassis materials should be changed to consider an old robot as a new one.

4.1.5. Use of IC engines is not allowed.

### 4.2. ROBOT DIMENSIONS

4.2.1. Dimensions of the robot should not exceed 30cm x 30cm x 30cm (l\*b\*h). (Tolerance of 5% is allowed in this regard).

4.2.2. Minimum track width of the robot (wheel to wheel) should be 25cm x 25cm.

### 4.3. POWER SUPPLY FOR ROBOTS

4.3.1. The potential difference between any two points should not exceed a limit of 12V and the robot should be powered only using batteries. (A tolerance of 5% is allowed).

4.3.2. If the robot suddenly stops in the arena due to any technical problem, you will be allowed to take a technical timeout. After taking one technical timeout, the timer will not be stopped if the robot suddenly stops in the arena due to any technical problem.

4.3.3. The power supply may be ON or OFF the board (i.e., one of the team members can hold the power supply). An external power supply will not be provided.

4.3.4. Changing the battery or battery polarity is not allowed during the race.

#### 4.4. WEIGHT

4.4.1. The maximum weight of the robot should not exceed 5kgs.

#### 5. SAFETY :

- 5.1. All the required accessories should be brought by the participants.
- 5.2. Team Roboveda is not responsible for any damage to the robot.
- 5.3. Technical assistance will not be provided by coordinators or technical deputies.

#### 6. THE COMPETITION AND MATCHES: There are two rounds

##### I. Qualifying Round

##### II. Final Round

#### 6.1. QUALIFYING ROUND

- 6.1.1. During the qualifying round, one team will be allowed to participate at a time.
- 6.1.2. The whole arena run should be completed within 15 minutes to qualify for the final round.
- 6.1.3. The winner of the round is judged by a circuit race of 1 lap.
- 6.1.4. If the robot touches any barricade once, a penalty (in seconds) will be levied, the specifications of which will be mentioned on the day of the event.
- 6.1.5. The total time taken by the participants to reach the finish line with penalties will be counted.
- 6.1.6. Arena will be explained to the participant before the race commences.
- 6.1.7. The width of the track will be in accordance with the robot specifications for the qualifying round. It may change in further rounds.

#### 6.2. FINAL ROUND

- 6.2.1. The top 12 teams that cracked the qualifying round in less than the total time (with penalties) of 15 minutes will be allowed for the final round.
- 6.2.2. Only one team is allowed to participate at a time.
- 6.2.3. The winner is judged by a circuit race of 1 lap.
- 6.2.4. If the robot touches the barricade once, a penalty (in seconds) will be given, the specifications of which will be mentioned on the day of the event.
- 6.2.5. Total time taken by participants to reach the finish line with penalties will be counted.
- 6.2.6. The team which cracks the arena in the least amount of time (with penalties) will be pronounced as the winner of the final round.

6.2.7. The robot participating in the final round should be the same one that participated in the qualifying round. Modifications will be allowed for the bot in between the rounds only if the following conditions are satisfied :

- The participant must show both the damaged part of the robot (wheels, chassis, motors) and the replaced part of the robot (wheels, chassis, motors) on the day of the final round.
- Heads or coordinators of the event will allow the modified robot only if the damaged part and replacement part are the same/similar.

## 7. COMPETITION RULES:

### 7.1. TECHNICAL TIMEOUTS

- 7.1.1. A technical timeout of 45 seconds can be taken by a team.
- 7.1.2. After the finishing of technical timeout, the timer will start again and will not be stopped under any circumstances.
- 7.1.3. No technical assistance will be provided by the event coordinators or heads during the event.

### 7.2. SCORING & PENALTIES

- 7.2.1. The robots that have cracked round-1 in the least amount of time in the qualifying round will be considered, and those teams will be allowed to participate in the final round.
- 7.2.2. If the robot touches any barricade once, a penalty (in seconds) will be given, the specifications of which will be mentioned on the day of the event.
- 7.2.3. The time taken by the robot to complete the race track, along with the penalties, is the total time taken.
- 7.2.4. Scoring schema (including penalties) will be disclosed on event day by heads and coordinators.

## 8. DISQUALIFICATION:

- 8.1. If a participant does not comply with the rules of the event, the robot will be disqualified from the event.
- 8.2. Robots should not damage any part of the arena. If the robot damages any part of the arena, it will be disqualified.
- 8.3. Misbehavior of any kind will not be tolerated and the team will be subjected to disqualification from Roboveda.
- 8.4. At any point of time during the match voltage should not be varied and should be constant, variation in voltage leads to disqualification.

## 9. COMPETITORS AND BEHAVIORAL RULES:

- 9.1. All participants will be given a Certificate of Participation.
- 9.2. Winners and runners will be given a Certificate of Merit.
- 9.3. All participants must bring both a college ID card and an event ID card.
- 9.4. Participants are allowed to participate once again if they satisfy rule 4.1.4
- 9.5. Participants are not allowed to use their own Stop watches.
- 9.6. Team Roboveda is not responsible for any kind of damage to your robot.
- 9.7. We request the participants not to assume anything that is not mentioned in the document without contacting the Coordinators or Technical Deputies.
- 9.8. It's our sincere request to all the participants to go through this rule book thoroughly and not a single point mentioned in the rule book will be compromised under any circumstances. We also request you have a copy of this rule book handy during the event.
- 9.9. In case of any discrepancies, the decision of co-ordinators and heads shall be final and no further argument will be entertained.
- 9.10. Make sure you visit our official website on a daily basis so that you will get to know if there are any changes made in the event.
- 9.11. Any kind of misbehavior will not be tolerated and will lead to disqualification of the team.
- 9.12. If you choose our hospitality, the payment must be made on the day you check-in.
- 9.13. In unforeseen circumstances, Team Roboveda abides by the decision of the Institution regarding the conductance of the event. Only registered participants will be informed about the change in event dates or cancellations. In that case, the registration fee will be refunded.

### **EVENT COORDINATORS:**

M. SAI VARUN: +919110347871  
P. BHAGYASREE: +916309247780

### **EVENT HEADS:**

M. SARVANAN: +919515632696  
K. TEJASHWINI: +919392423513

